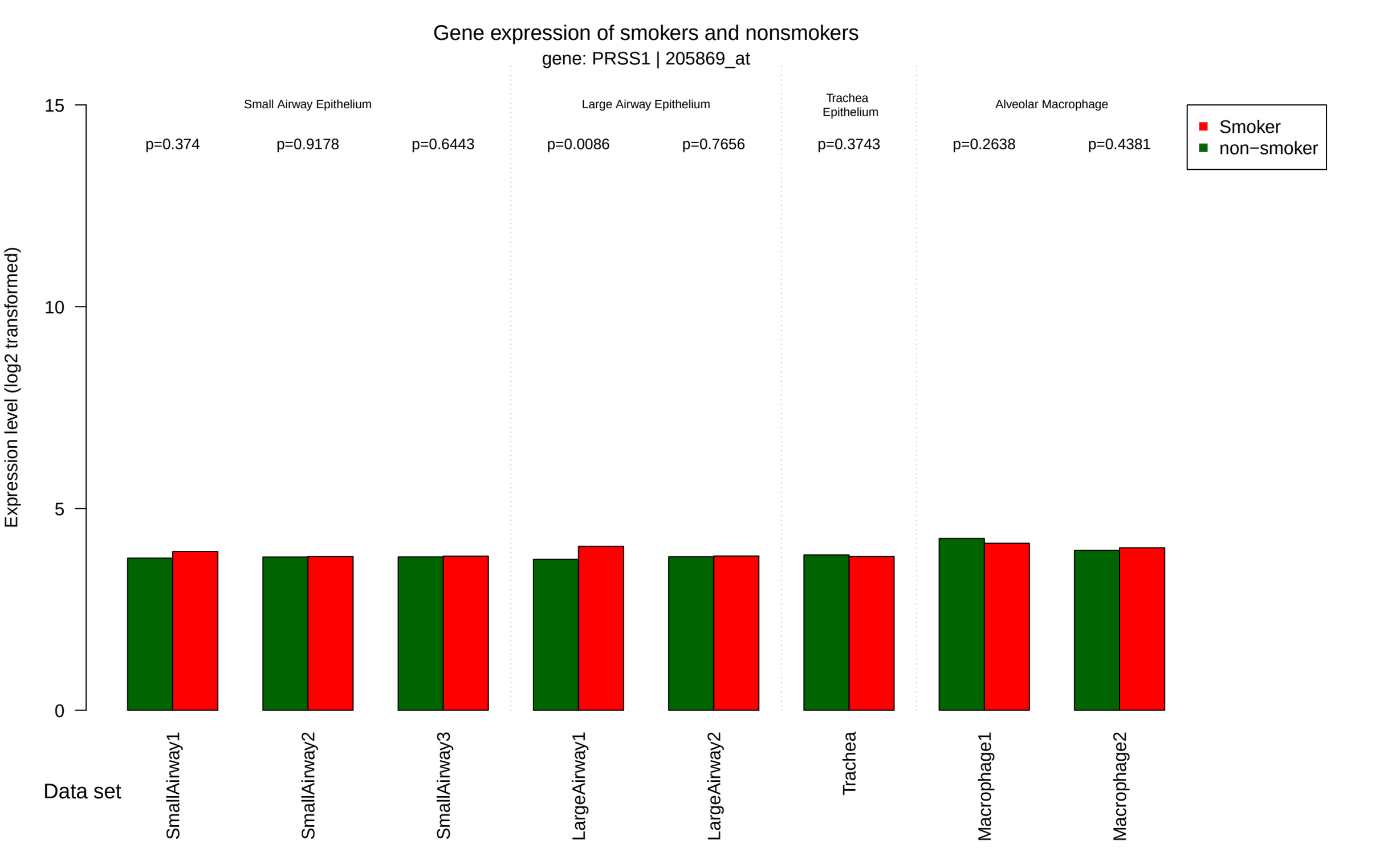
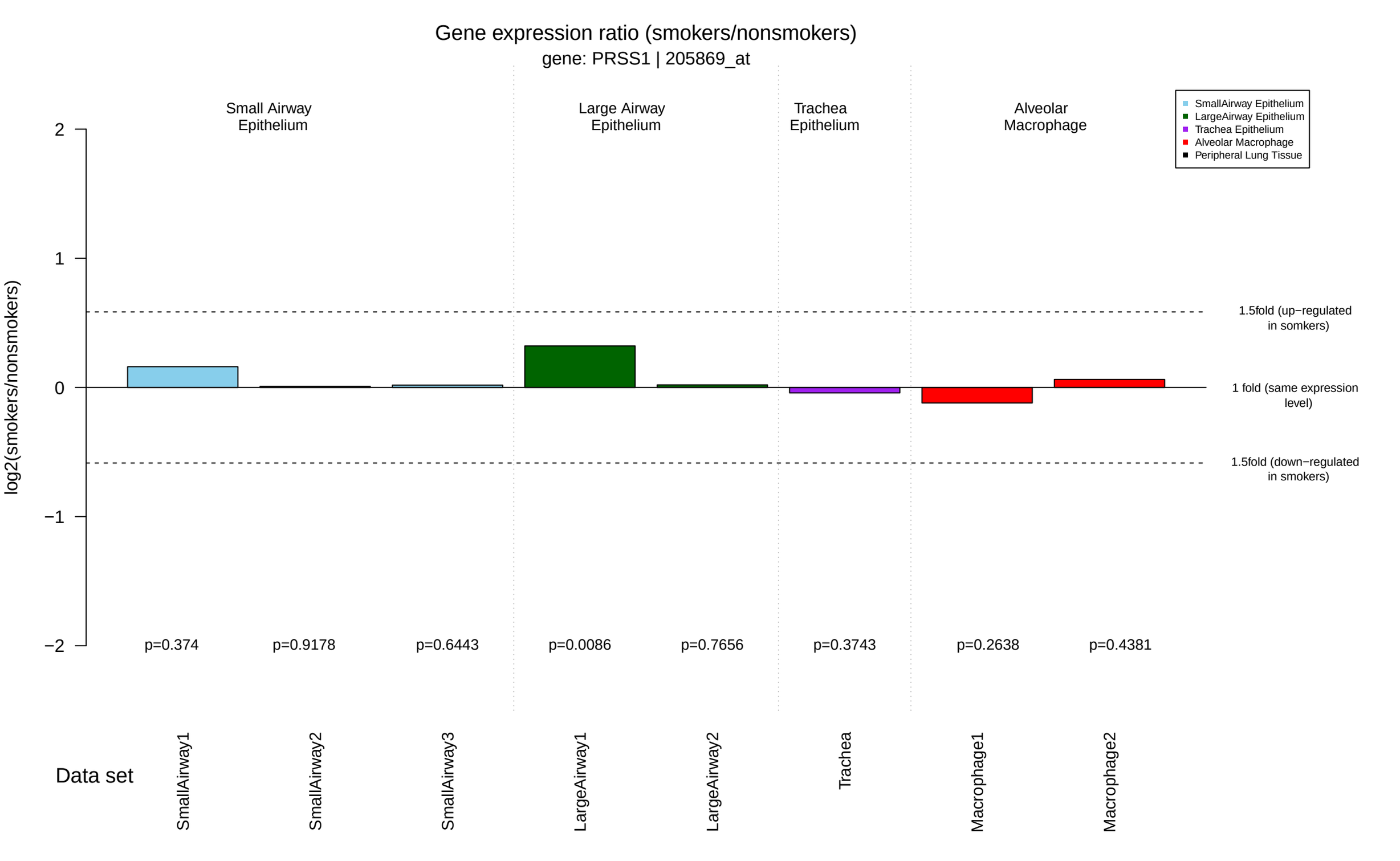
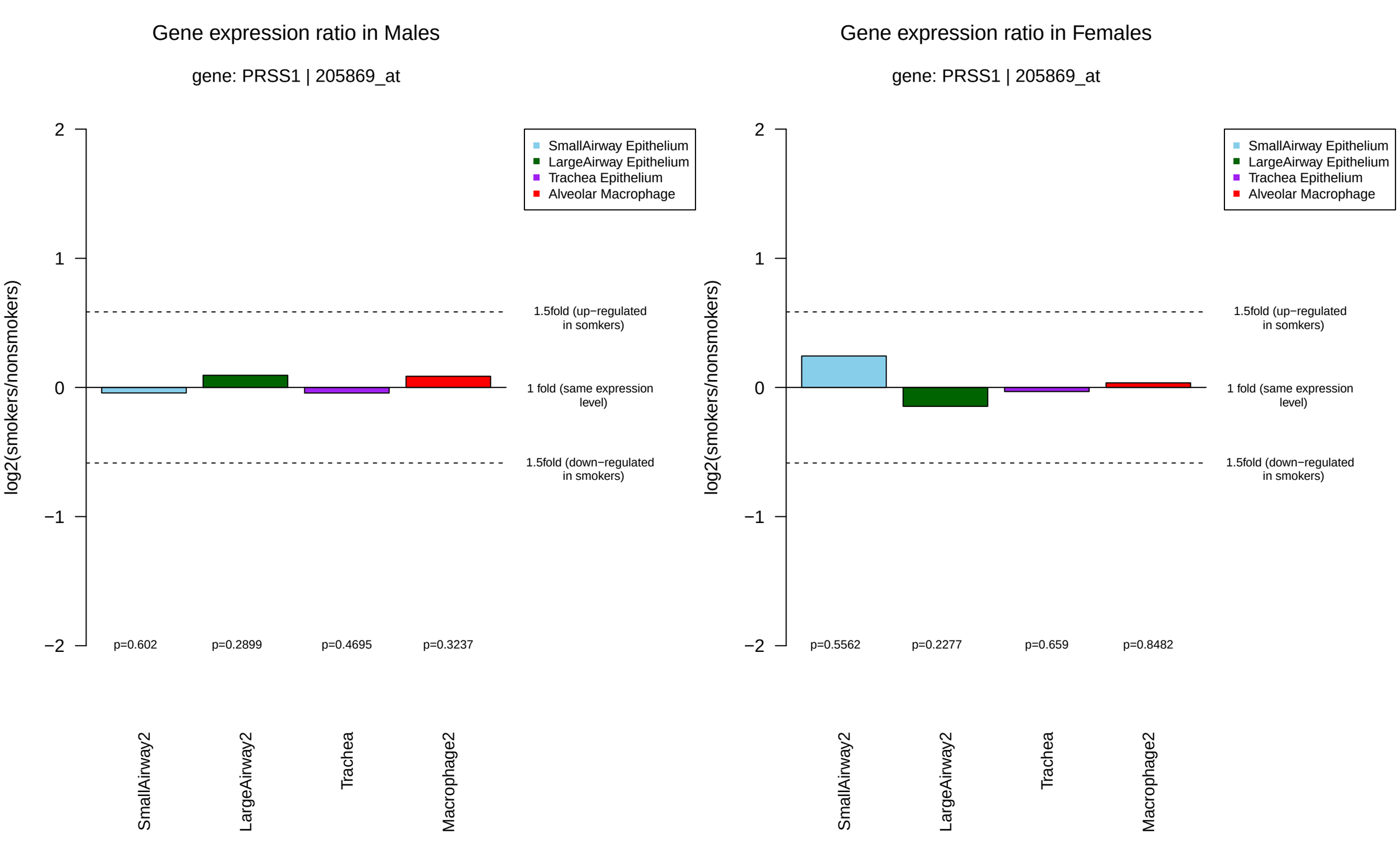
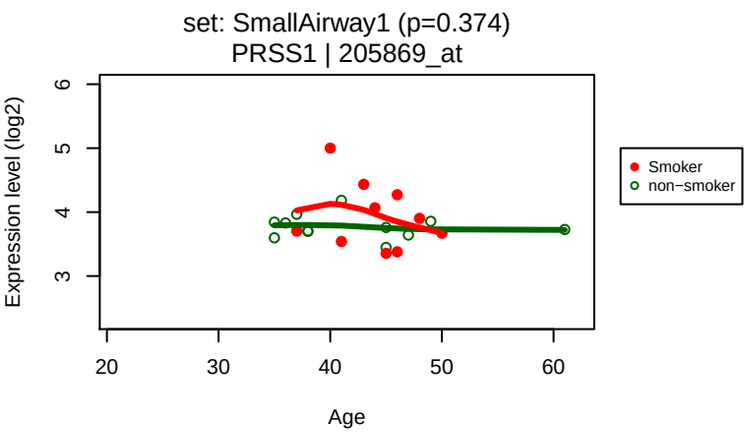
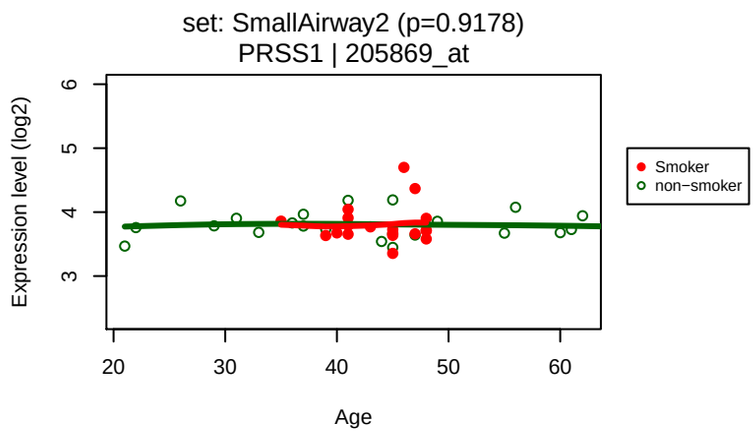
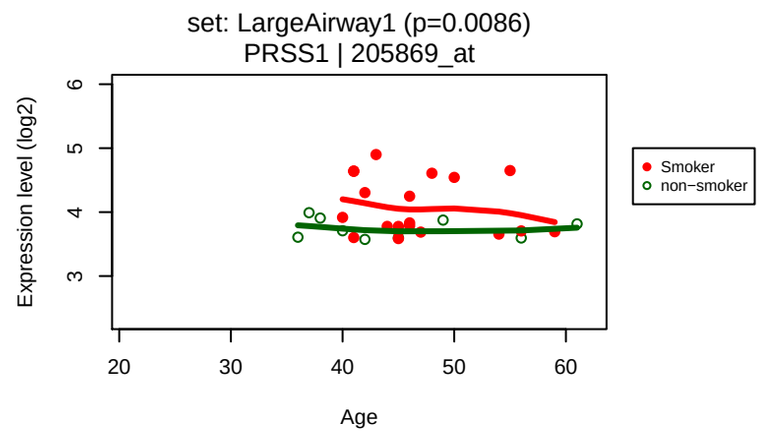

Gene expression of smokers and nonsmokers
gene: PRSS1 | 205869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.374
p=0.9178
p=0.6443
p=0.0086
p=0.7656
p=0.3743
p=0.2638
p=0.4381
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

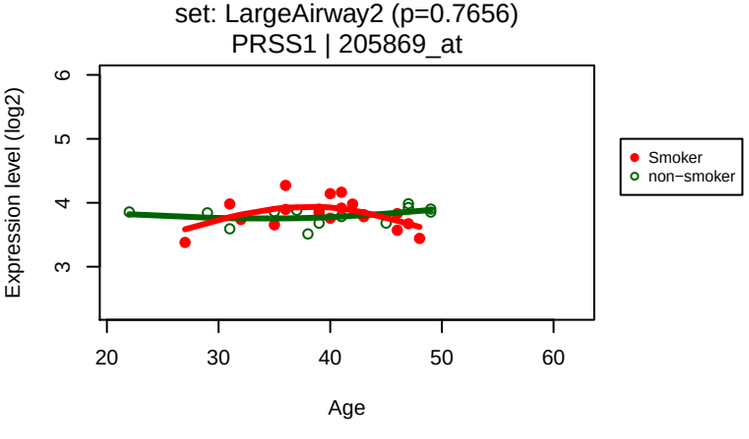

Gene expression ratio (smokers/nonsmokers)
gene: PRSS1 | 205869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.374
p=0.9178
p=0.6443
p=0.0086
p=0.7656
p=0.3743
p=0.2638
p=0.4381
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

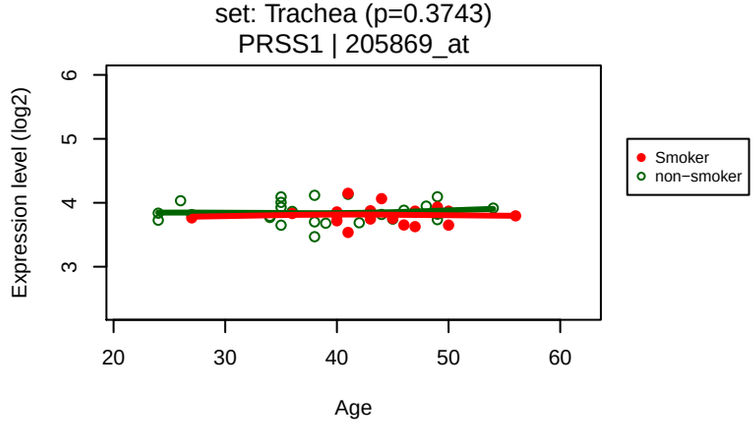

Gene expression ratio in Males
gene: PRSS1 | 205869_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.602
p=0.2899
p=0.4695
p=0.3237
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS1 | 205869_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5562
p=0.2277
p=0.659
p=0.8482
SmallAirway2
LargeAirway2
Trachea
Macrophage2

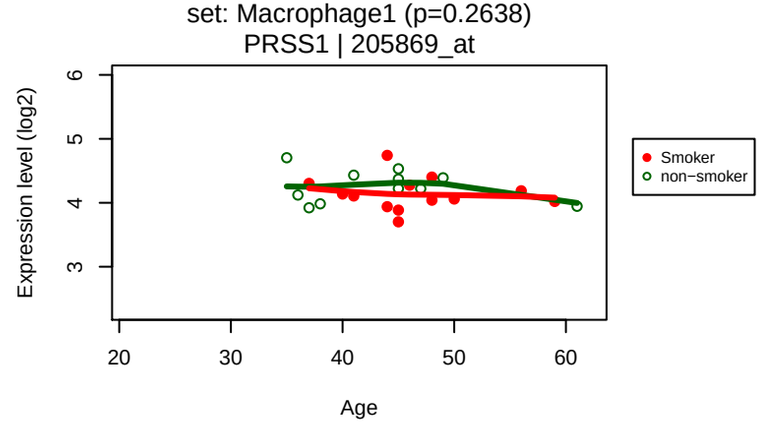

set: SmallAirway1 (p=0.374)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker

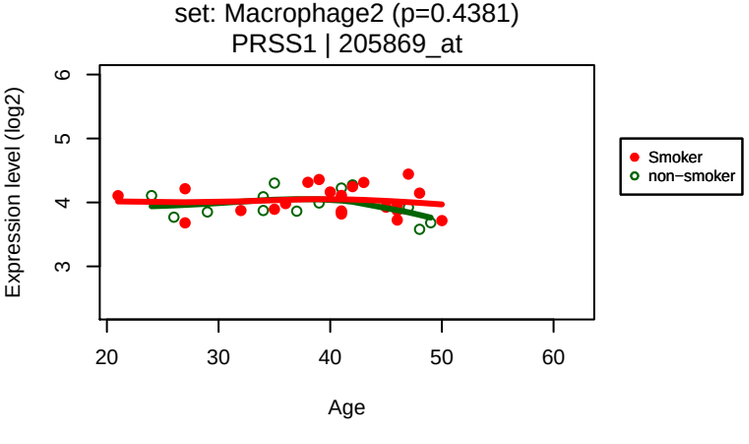

set: SmallAirway2 (p=0.9178)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker

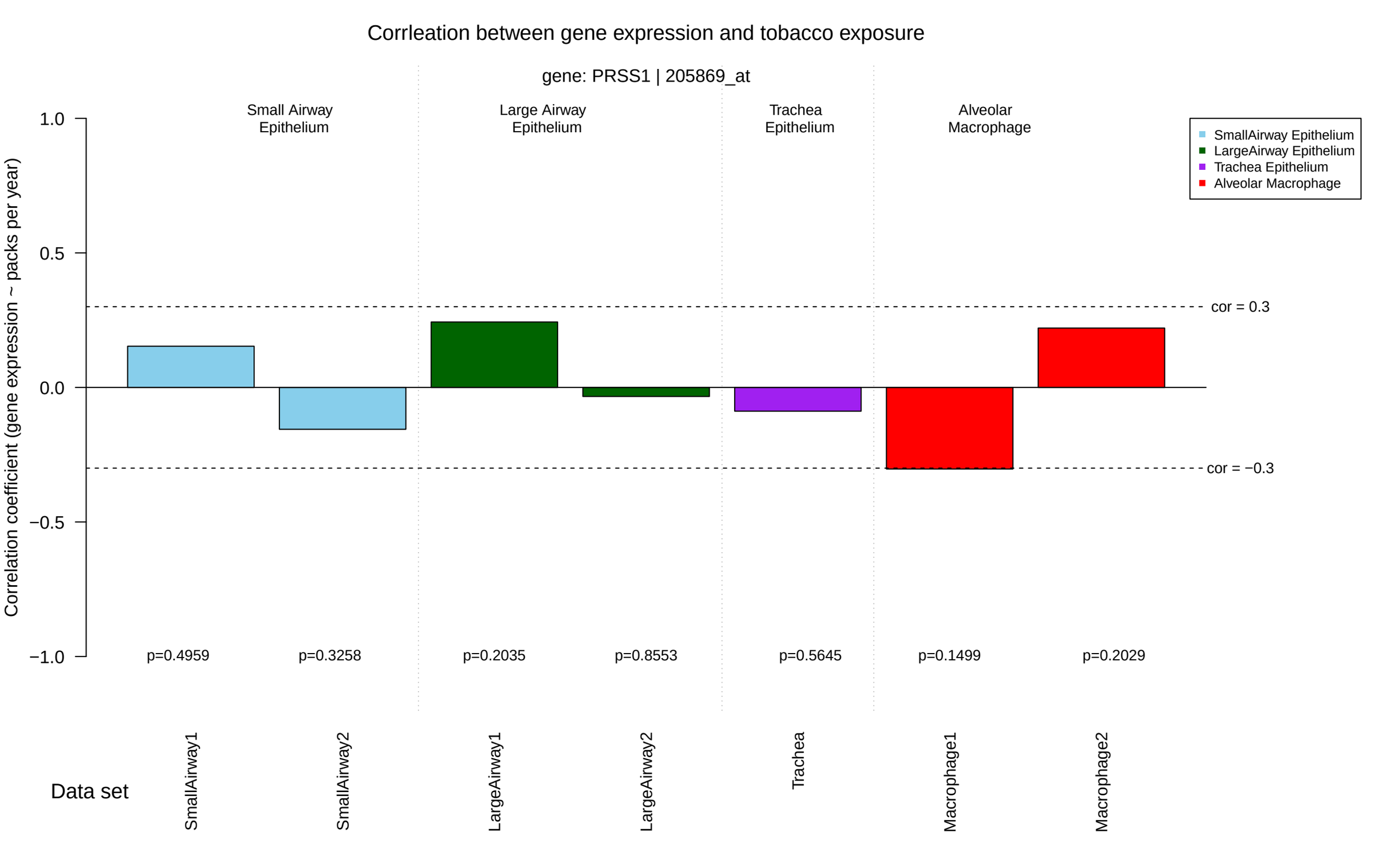

set: LargeAirway1 (p=0.0086)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7656)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3743)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2638)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4381)
PRSS1 | 205869_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS1 | 205869_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4959
p=0.3258
p=0.2035
p=0.8553
p=0.5645
p=0.1499
p=0.2029
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2